

CS-302: Artificial Intelligence

LECTURE NO 9:

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Organization of Lecture

1. Single-State Problem Formulation V/S Multiple-State Problem Formulation.
2. Default Reasoning.
3. Monotonic Reasoning V/S Non-Monotonic Reasoning.
4. Production System in A.I.

1. Single-State Problem Formulation V/S Multiple-State Problem Formulation (1/2)

1) Single-State Problem formulation V/S Multiple-state Problem formulation

Single-State Problem formulation

↳ Exact Prediction is possible

State → is exactly known after any action sequence.

Accessibility → Information achieved through sensors.

Consequences → of actions are known to agent.

Multiple-State Problem formulation

Semi-exact prediction is possible

State → not exactly known, but limited to a set of possible states.

Accessibility → not all essential information can be achieved through sensors.
[Reasoning]

Consequences → Not always known to the agents [Randomness]

1. Single-State Problem Formulation V/S Multiple-State Problem Formulation (2/2)

Goal → unique for each known initial state.	Goal → No fixed action sequence that leads to Goal
↳ [Simple but restricted] → e.g.: Vacuum World	↳ Less restricted but more complex. → e.g.: Vacuum world but no sensors.
↳ It can't deal with incomplete accessibility [Limitation] ↳ Incomplete knowledge about the consequences can alter the world [Limitation]	↳ Can't deal with changes in the world during execution [Limitation]

2. Default Reasoning (1/2)

2) Default Reasoning

↳ It is very common form of non-monotonic reasoning

↳ Conclusions are drawn based on what is most likely to be true.

↳ Approaches to Default Reasoning → (Non-Monotonic Logic and Default Logic)

2. Default Reasoning (2/2)

Non-Monotonic Logic

↳ Truth of Proposition may be changed when new information are added and logic may be built to allow the statement to be retracted.

↳ Modal operator (M) → Consistent with everything we know

Examples

→ Fact 'F' is true, we attempt to prove $\neg F$ ^{can}

↳ Fail $\Rightarrow F$ is True

For all x

↳ If x plays an instrument

→ $\forall x: \text{plays-instrument}(x) \wedge$ and

$M \text{ manages}(x) \rightarrow \text{jazz-musician}(x)$

x manages it consistently

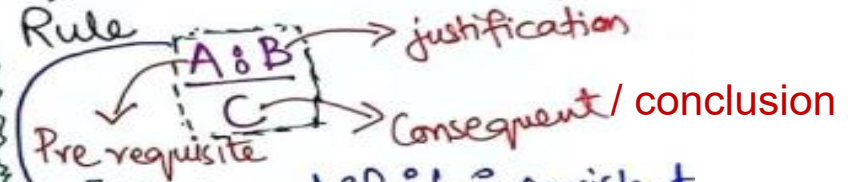
then

Conclusion

x is a jazz musician

Default Logic

↳ Initiates a new Inference Rule



[If A and if it is consistent with rest of what is known to assume that B, then conclude that C.]

Examples

$\text{BIRD}(X) : \text{FLY}(X)$

$\text{FLY}(X)$

$\text{BIRD}(\text{Tweety}) : \text{FLIES}(\text{Tweety})$

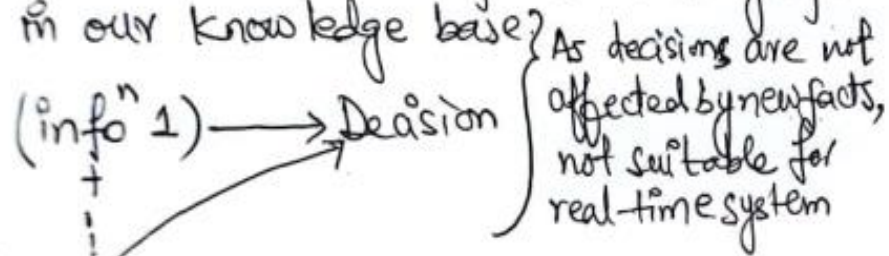
$\text{FLIES}(\text{Tweety})$

if tweety is a bird and it is consistent that tweety can fly then tweety can fly.

3. Monotonic Reasoning V/S Non-Monotonic Reasoning

3) Monotonic Reasoning V/S Non-Monotonic Reasoning

Monotonic Reasoning: Once the conclusion is taken, then it will remain same even if we add some other infoⁿ to existing infoⁿ in our knowledge base.



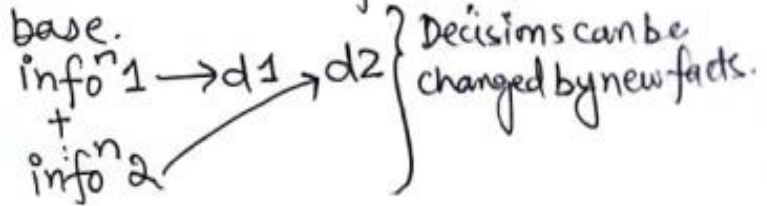
E.g.:
→ Earth revolves around SUN] earth is not round]
New infoⁿ

→ All old proofs are valid.] Adv

→ Can't represent real-world scenarios] Disadv

→ New knowledge from real-world can't be added] Disadv.

Non-Monotonic Reasoning: In this some conclusion may be invalidated if we add some more infoⁿ to our knowledge base.



→ Helpful in Real-world scenarios.] Adv.

E.g.: Birds can fly
Penguins can't fly] Alex can fly
Alex is a bird
+ Alex is a Penguin] Alex can't fly

Disadv.

→ Can't be used for theorem proving.

4. Production System in A.I (1/3)

4) Production System in A.I

- ↳ Helps in structuring A.I Programs in a way that facilitates describing and Performing the search process.
- ↳ Production System consists of 4 parts:
 - (i) Set of rules (if-then etc)
 - (ii) Knowledge Base (Data base)
 - (iii) Control Strategy (It specifies the order in which rules need to be compared to the data base so that the conflict is resolved in a minimum time).
 - (iv) Rule Applier (Rules are applied upon control strategy).

4. Production System in A.I (2/3)

Steps to solve the Problem

- i) First reduced the problem so that it can be shown in precise statement.
- ii) Problem can be solved by searching a path through space. [Start State $\longrightarrow$ Goal State].
- iii) Solving process can be modelled through a production system.

Advantages

- i) Structuring A.I Problems] Excellent Tool.
- ii) Highly Modular] rules can be added, removed or changed.
- iii) Rules are expressed in natural form] Rule is easy to understand.

4. Production System in A.I (3/3)

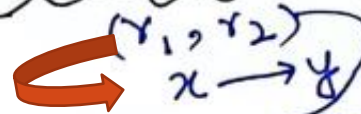
Characteristics:

✓ i) Monotonic Production System : Application of a rule never prevents later application of another rule] Rules are independent

ii) Non-Monotonic Production System which is not true

✓ iii) Partially Commutative Production System

Allowable



$r_1 = \text{rule 1}$
 $r_2 = \text{rule 2}$

permutation (Any combination of rules r_1 & r_2)
 $\rightarrow (x \rightarrow y)$

iv) Commutative Production System

↳ It is both monotonic and partially commutative production system.

References: Textbooks

1. *“Artificial Intelligence”, by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
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3. *“Computational Intelligence: A logical Approach”, by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *“Artificial Intelligence: Structures and Strategies for Complex Problem Solving”, by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *“AI: A new Synthesis”, by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *“Artificial Intelligence: Theory and Practice” by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*